

# Evaluation of Clustering Algorithms

## How Do We Know If a Clustering Is Good?

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Machine Learning Lecture

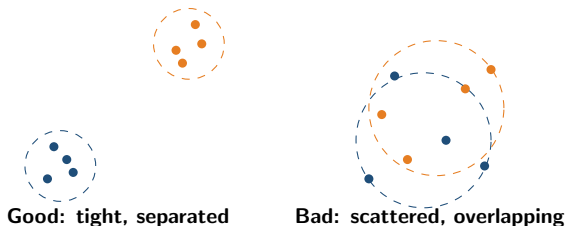
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# What We Will Learn Today

- ① What is clustering evaluation and why we need it
- ② Why train/test split does NOT work here (key idea)
- ③ Related terms: cohesion, separation, ground truth
- ④ Two families: internal vs external evaluation
- ⑤ Architecture: the evaluation process step by step
- ⑥ Numerical example: computing Silhouette Score by hand
- ⑦ Elbow Method: how and why it is used to choose  $k$

# What is Clustering Evaluation?

- Clustering groups similar data points together, with **no labels given**.
- After clustering, we must ask: **is this grouping actually good?**
- **Clustering evaluation** is the set of methods used to measure the quality of the groups a clustering algorithm produced.



- Why used: to **compare algorithms**, to **choose the right number of clusters ( $k$ )**, and to know if the result is **trustworthy** before using it in a business decision.

# Why NOT Train/Test Split? (Key Idea)

## In supervised learning

We split data so we can check predictions against **true labels** the model never saw.

Train set (learn)    +    Test set (has known answers to check against)

## In unsupervised learning (clustering)

**There are no labels at all**, not even in a held-out set. Nothing exists to “check the answer” against, so a train/test split gives us nothing to compare the test predictions to.

# Why NOT Train/Test Split? (Continued)

Supervised: Train on 80%  
Test accuracy on 20% **labels**

Unsupervised: Train on 80%  
Test on 20% but... **no labels to check!**

- Splitting only reduces how much data the clustering algorithm can see; there is no ground truth to test predictions against, so it does not test generalization the way it does in supervised learning.
- Instead, clustering is evaluated using:
  - 1 **Internal metrics** — use only the data and the cluster assignments (no labels needed)
  - 2 **External metrics** — used only when true labels DO exist, to check clustering against them (not for splitting into train/test)

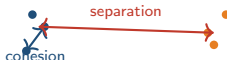
# Related Terms 1: Cohesion and Separation

## Cohesion (Intra-cluster distance)

How close points are to **each other inside the same cluster**. Lower is better (tighter cluster).

## Separation (Inter-cluster distance)

How far apart **different clusters** are from each other. Higher is better (clearer boundaries).



- A good clustering has **low cohesion distance** (tight groups) and **high separation distance** (well apart groups).

## Related Terms 2: Ground Truth and WCSS

- **Ground truth labels:** the real category of each point, if it happens to be known (e.g. from a labeled benchmark dataset). Normal real-world clustering does NOT have this.
- **WCSS (Within-Cluster Sum of Squares):** total squared distance from each point to its own cluster's center.

$$\text{WCSS} = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

- Used by the **Elbow Method** to help pick the number of clusters  $k$ .
- **Centroid ( $\mu_i$ ):** the average position of all points inside cluster  $i$ .

# Two Families of Evaluation

## Internal Evaluation

Uses only the data and cluster assignments. No true labels needed.

**Used almost always**, since real clustering has no labels.

- Silhouette Score
- Davies-Bouldin Index
- WCSS / Elbow Method
- Calinski-Harabasz Index

## External Evaluation

Compares clusters to **known true labels**. Only possible when labels exist (e.g. testing a new algorithm on a labeled benchmark).

- Rand Index (RI)
- Adjusted Rand Index (ARI)
- Normalized Mutual Information (NMI)
- Purity

In real projects, clustering has no ground truth, so internal metrics are the everyday tool.



# Silhouette Score: The Most Popular Internal Metric

For a single point  $i$ :

$a(i)$  = avg distance to other points in the SAME cluster (cohesion)

$b(i)$  = avg distance to points in the NEAREST OTHER cluster (separation)

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

| Silhouette value | Meaning                                          |
|------------------|--------------------------------------------------|
| close to +1      | point fits its own cluster well, far from others |
| close to 0       | point sits on the border between two clusters    |
| close to -1      | point is likely in the wrong cluster             |

Average  $s(i)$  over all points gives the overall **Silhouette Score** for the whole clustering.

# Other Internal Metrics

## Davies-Bouldin Index (DBI)

Average similarity between each cluster and its most similar other cluster. **Lower is better** (0 is the best possible score).

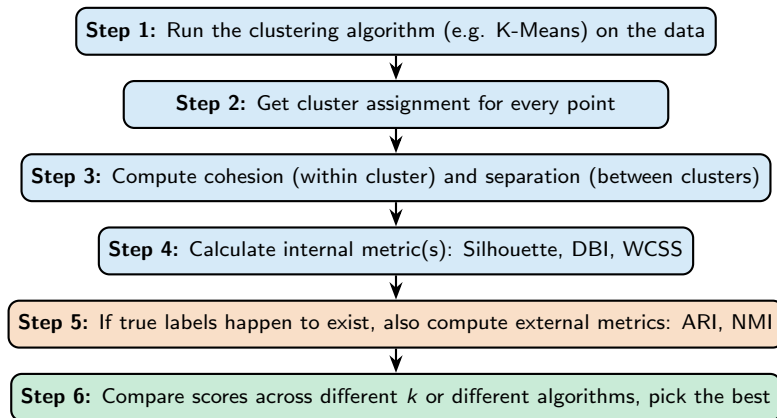
## Elbow Method (using WCSS)

Run clustering for  $k = 1, 2, 3, \dots$  and plot WCSS. Pick the  $k$  where the curve bends like an elbow, after which adding more clusters barely helps.

## Calinski-Harabasz Index

Ratio of between-cluster separation to within-cluster cohesion. **Higher is better.**

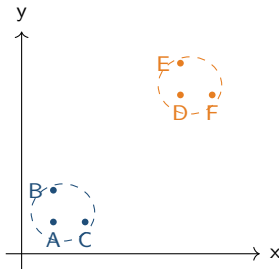
# Architecture: The Evaluation Process



# Numerical Example: The Dataset

Six points already clustered into two groups of three by K-Means:

| Pt | Coords | Cl. |
|----|--------|-----|
| A  | (1, 1) | 1   |
| B  | (1, 2) | 1   |
| C  | (2, 1) | 1   |
| D  | (5, 5) | 2   |
| E  | (5, 6) | 2   |
| F  | (6, 5) | 2   |



**Task:** build the pairwise distance table, then compute the Silhouette Score for every point.

# Step 1: Pairwise Distance Table

Euclidean distance  $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$  between every pair of points:

|   | A     | B     | C     | D     | E     | F     |
|---|-------|-------|-------|-------|-------|-------|
| A | 0     | 1.000 | 1.000 | 5.657 | 6.403 | 6.403 |
| B | 1.000 | 0     | 1.414 | 5.000 | 5.657 | 5.831 |
| C | 1.000 | 1.414 | 0     | 5.000 | 5.831 | 5.657 |
| D | 5.657 | 5.000 | 5.000 | 0     | 1.000 | 1.000 |
| E | 6.403 | 5.657 | 5.831 | 1.000 | 0     | 1.414 |
| F | 6.403 | 5.831 | 5.657 | 1.000 | 1.414 | 0     |

- Notice the pattern: distances **within** {A,B,C} and within {D,E,F} are small (around 1 to 1.4).
- Distances **between** the two groups are large (around 5.0 to 6.4).
- This table is all we need — every  $a(i)$  and  $b(i)$  below is just an average of numbers already in it.

## Step 2: Compute $a(i)$ and $b(i)$ for Cluster 1

**Cluster 1** = {A, B, C}.  $a(i)$  = avg distance to the other 2 points in Cluster 1.  $b(i)$  = avg distance to all 3 points in Cluster 2.

| Point | $a(i)$                            | $b(i)$                                    |
|-------|-----------------------------------|-------------------------------------------|
| A     | $\frac{1.000 + 1.000}{2} = 1.000$ | $\frac{5.657 + 6.403 + 6.403}{3} = 6.154$ |
| B     | $\frac{1.000 + 1.414}{2} = 1.207$ | $\frac{5.000 + 5.657 + 5.831}{3} = 5.496$ |
| C     | $\frac{1.000 + 1.414}{2} = 1.207$ | $\frac{5.000 + 5.831 + 5.657}{3} = 5.496$ |

Every number here is read directly off the distance table's A/B/C rows.

## Step 2 (contd.): Compute $a(i)$ and $b(i)$ for Cluster 2

**Cluster 2** = {D, E, F}.  $a(i)$  = avg distance to the other 2 points in Cluster 2.  $b(i)$  = avg distance to all 3 points in Cluster 1.

| Point | $a(i)$                            | $b(i)$                                    |
|-------|-----------------------------------|-------------------------------------------|
| D     | $\frac{1.000 + 1.000}{2} = 1.000$ | $\frac{5.657 + 5.000 + 5.000}{3} = 5.219$ |
| E     | $\frac{1.000 + 1.414}{2} = 1.207$ | $\frac{6.403 + 5.657 + 5.831}{3} = 5.964$ |
| F     | $\frac{1.000 + 1.414}{2} = 1.207$ | $\frac{6.403 + 5.831 + 5.657}{3} = 5.964$ |

Same idea, just reading the D/E/F rows of the table instead.

## Step 3: Silhouette Score for Every Point

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

| Point | $a(i)$ | $b(i)$ | $s(i)$                                |
|-------|--------|--------|---------------------------------------|
| A     | 1.000  | 6.154  | $\frac{6.154 - 1.000}{6.154} = 0.837$ |
| B     | 1.207  | 5.496  | $\frac{5.496 - 1.207}{5.496} = 0.780$ |
| C     | 1.207  | 5.496  | $\frac{5.496 - 1.207}{5.496} = 0.780$ |
| D     | 1.000  | 5.219  | $\frac{5.219 - 1.000}{5.219} = 0.808$ |
| E     | 1.207  | 5.964  | $\frac{5.964 - 1.207}{5.964} = 0.798$ |
| F     | 1.207  | 5.964  | $\frac{5.964 - 1.207}{5.964} = 0.798$ |

Every point scores close to +1: each one sits much closer to its own cluster than to the other one



# Overall Silhouette Score

Average the six  $s(i)$  values to get the score for the **whole clustering**:

$$\bar{s} = \frac{0.837 + 0.780 + 0.780 + 0.808 + 0.798 + 0.798}{6} = \frac{4.801}{6} \approx \mathbf{0.800}$$

**0.800** falls in the “strong clustering structure” range  
(both groups of three are tight internally and clearly separated  
from each other)

- A and D score highest (0.837, 0.808): each is the “corner” point closest to its own two cluster-mates and furthest from the other group.
- B, C, E, F score slightly lower: each sits a bit further from one of its own cluster-mates, but still clearly separated from the other cluster.

# What Does Each Score Value Mean?

**Per-point  $s(i)$ , range  $-1$  to  $+1$ :**

| Value         | Meaning                                                       |
|---------------|---------------------------------------------------------------|
| close to $+1$ | fits its own cluster well, far from the nearest other cluster |
| close to $0$  | sits right on the border between two clusters                 |
| close to $-1$ | likely placed in the wrong cluster                            |

**Overall average Silhouette Score, used to judge or compare clusterings:**

| Range       | Interpretation                              |
|-------------|---------------------------------------------|
| 0.71 – 1.00 | Strong, well separated clustering structure |
| 0.51 – 0.70 | Reasonable structure, usable                |
| 0.26 – 0.50 | Weak structure, try another $k$ or method   |
| below 0.25  | No substantial clustering structure found   |

Our example scored **0.800** — strong structure, matching the two tight and well separated groups of three seen in the figure.

# The Elbow Method: What It Is

- Algorithms like **K-Means** need the number of clusters  $k$  decided **before** they run.
- The Elbow Method is a technique to **pick a good value of  $k$**  using WCSS (Within-Cluster Sum of Squares), the same cohesion measure seen earlier.

## Core idea

As  $k$  increases, WCSS always keeps going down (more clusters can only fit the data tighter). The Elbow Method looks for the point where adding more clusters **stops giving a meaningful improvement**.

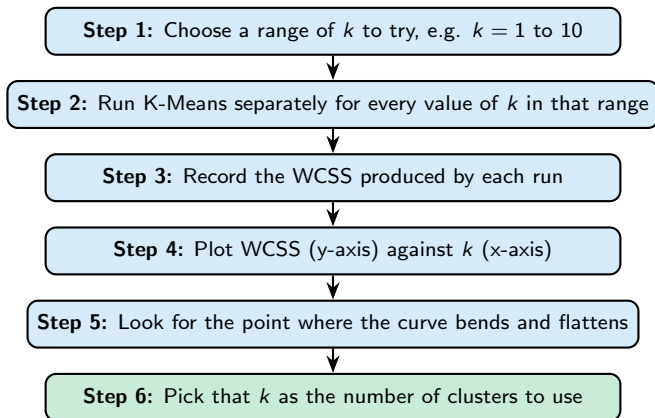
- That bend in the curve looks like an elbow, hence the name.

# Why Is It Used?

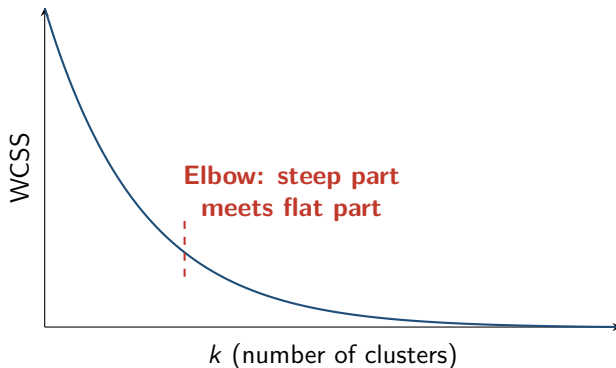
- Clustering has **no labels**, so there is no automatic way to know the “correct” number of groups.
- Choosing  $k$  badly causes real problems:
  - **Too small**  $k$ : distinct groups get merged together, hiding real structure
  - **Too large**  $k$ : natural groups get split apart, and clusters start fitting noise instead of real patterns
- The Elbow Method gives a **simple, visual, repeatable way** to justify a choice of  $k$  before doing any deeper analysis.

It does not tell you the “true” number of clusters — it estimates a reasonable, defensible one from the data’s own behavior.

# How It Is Used: The Process



# Reading the Elbow Shape



- **Steep drop (small  $k$ ):** each new cluster still captures a lot of real structure.
- **Flattening (larger  $k$ ):** extra clusters mostly just split existing groups, WCSS barely improves.
- The  $k$  at the bend is the practical trade-off between simplicity and fit.

# Strengths and Limitations of the Elbow Method

## Strengths

- Simple to compute and to explain
- Works with any algorithm that produces WCSS-like output
- Good first check before deeper analysis

## Limitations

- The bend can be unclear or gradual on real data
- Reading the plot is somewhat subjective
- Best combined with Silhouette Score or DBI to confirm the choice of  $k$

# Summary

- 1 Clustering evaluation checks the **quality of groups** formed with no labels.
- 2 **Train/test split does not apply**: there is no ground truth to test predictions against in unsupervised learning.
- 3 **Internal metrics** (Silhouette, DBI, WCSS/Elbow) use only the data itself and are used almost always.
- 4 **External metrics** (ARI, NMI, Purity) are used only on the rare occasion true labels exist.
- 5 Silhouette Score combines **cohesion** and **separation** into one number between  $-1$  and  $+1$ .
- 6 Elbow Method plots WCSS against  $k$  and picks the bend point.

**Practice:** add a 5th point  $E(3,3)$  to Cluster 2 and recompute the overall Silhouette Score.